FYUGP B.Sc. PHYSICS HONOURS

PROPERTIES OF SOLIDS

Semester V · Major Elective (Materials Science)

MODULE 1: CRYSTAL STRUCTURE

Past questions

Section A: 1 to 3 marks

Crystal lattice, unit cell and basis

1	The process of determination of crystal structure is called _____. **(1 mark)**	*March 2022*
2	Define crystal lattice. **(2 marks)**	*March 2022*
3	What is meant by unit cell ? Give an example. **(2 marks)**	*March 2025*
4	Explain the terms crystal lattice and basis. **(2 marks)**	*March 2025*
5	What is a unit cell, and how does it differ from a primitive cell ? **(2 marks)**	*April 2026*

Symmetry operations, point groups and space groups

6	Define crystal symmetry and give an example. **(2 marks)**	*April 2026*

Lattice directions, planes, Miller indices and interplanar spacing

7	Give the expression for lattice constant ‘a’ of a cubic system. **(1 mark)**	*March 2023*
8	What are miller indices ? **(2 marks)**	*March 2023*
9	Indicate a (111) direction in a cubic crystal. **(2 marks)**	*March 2023*
10	What is the importance of Miller indices in crystallography ? **(2 marks)**	*April 2026*

Simple crystal structures

11	Atomic packing factor of a diamond cubic structure is _____. **(1 mark)**	*March 2020*
12	Give an example for simple cubic structure. **(1 mark)**	*March 2020*
13	What is meant by atomic packing factor ? **(2 marks)**	*March 2023*
14	What do you mean by the co-ordination number of a lattice ? What is the co-ordination number of an fcc lattice ? **(2 marks)**	*March 2023*
15	Draw the atomic positions in the cubic cell of diamond projected on a face. **(2 marks)**	*March 2023*
16	What is meant by atomic packing factor (APF) ? **(2 marks)**	*April 2026*

X-ray diffraction, Bragg’s law and reciprocal lattice

17	Write down Bragg’s equation. **(2 marks)**	*March 2022*
18	Vector form of Bragg’s law is _____. **(1 mark)**	*March 2023*

Section B: 4 to 5 marks

Crystal lattice, unit cell and basis

19	Define the terms : lattice, basis, crystal structure and unit cell. **(4 marks)**	*March 2020*

Symmetry operations, point groups and space groups

20	What are symmetry elements ? **(4 marks)**	*March 2023*

Bravais lattices and crystal systems

21	Write a brief note on crystal systems. **(4 marks)**	*March 2022*

Lattice directions, planes, Miller indices and interplanar spacing

22	Calculate the inter planar spacing for (321) plane in simple cubic lattice with inter atomic spacing a = 4.21 Å. **(4 marks)**	*March 2020*
23	Copper has fee structure with the lattice constant 0.361 nm. Calculate the interplanar spacing for (112) and (120) planes. **(4 marks)**	*March 2022*
24	Find the miller indices of a plane that makes intercepts as a, b and c axes equal to 3 Å, 4 Å and 3 Å in a tetragonal crystal with c/a ratio as 1.5. **(4 marks)**	*March 2023*
25	Find the lattice spacing of a cubic crystal with lattice parameter 4.2 Å for (1, 1, 1) planes. **(5 marks)**	*April 2026*

Simple crystal structures

26	Show that the packing factor of bcc lattice is $\pi\sqrt{3}/8$. **(4 marks)**	*March 2020*
27	Derive the expression for volume of unit cell in body centred cubic structure. **(4 marks)**	*March 2020*
28	Find out the packing fraction of FCC. **(5 marks)**	*March 2025*

X-ray diffraction, Bragg’s law and reciprocal lattice

29	Explain how X- rays are used for determining the crystal structure. **(4 marks)**	*March 2020*

30	A monochromatic beam of X ray of λ = 0.7 Å undergoes first order Bragg reflection from the plane (3 0 2) of a cubic crystal at a glancing angle of 35°. Determine the lattice constant [symbol missing in print]. **(5 marks)**	*March 2023*
31	Derive and explain Bragg's law. **(5 marks)**	*March 2025*
32	Explain how Bragg's X-ray spectrometer works and its application. **(5 marks)**	*April 2026*
33	A monochromatic X-ray beam of wavelength 1.54 Å is used in a powder diffraction experiment. If the first-order diffraction peak is observed at a Bragg angle of 30°, calculate the interplanar spacing (d) of the crystal. **(5 marks)**	*April 2026*

Section C: 10 marks

Bravais lattices and crystal systems

34	Explain the crystal system in detail with examples. **(10 marks)**	*March 2020*

Simple crystal structures

35	Explain the close-packed structures in crystal. **(10 marks)**	*March 2022*
36	Explain the atomic arrangements in an hcp structure and obtain the corresponding packing fraction. **(10 marks)**	*March 2023*

X-ray diffraction, Bragg's law and reciprocal lattice

37	Explain Bragg's law and Bragg's X-ray spectrometer. **(10 marks)**	*March 2020*